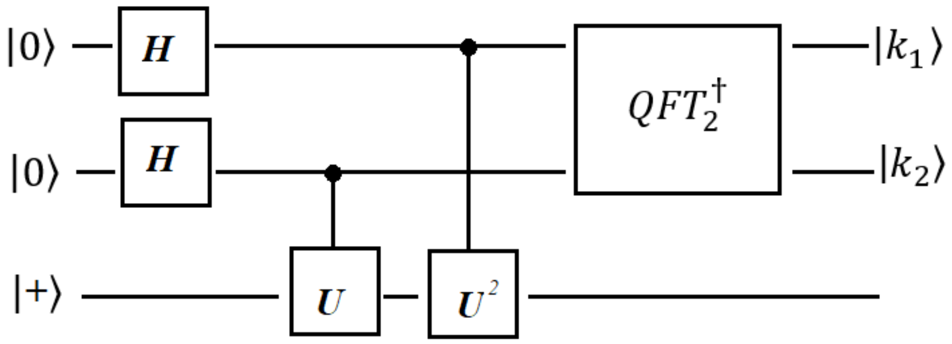
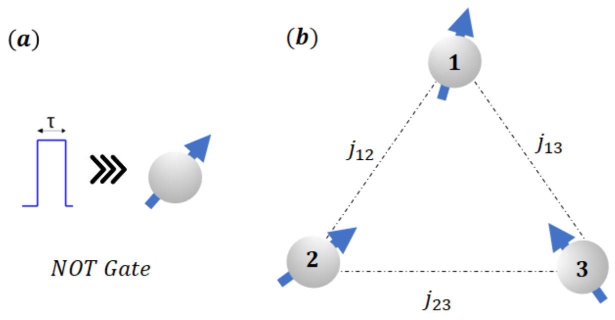
ELK336-Quantum Computation
FINAL-EXAM – (Take Home)
DUE 05.06.2023 (23.00)

1- (a) Given the operator $U = R_x(\pi)$, show that initial state $|+\rangle$ is an eigenstate of U.
Then obtain the phase factor ϕ in the binary form by use of the corresponding eigenvalue $\lambda = e^{2\pi i\phi}$.
Finally show that $|\phi\rangle = |k_1 k_2\rangle = |11\rangle$.
(b) Find the output $|k_1 k_2\rangle$ of the algorithm given below where U is the R_x rotation given above.
Draw the whole algorithm including $QFT_2^\dagger$ and obtain $|k_1 k_2\rangle$ step by step (in Dirac notation) starting from $|00+\rangle$.



2- The density matrix of a pure quantum state $|\psi\rangle$ is defined as $\rho = |\psi\rangle\langle\psi|$and the density matrix of the collection of pure states $|\psi_i\rangle$ with the corresponding probabilities p_i is defined as
$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$.
(a) Find a density matrix of $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and show that this density matrix corresponds a pure state (by using the purity).
(b) Find a density matrix composed of quantum states $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$,
$|\psi_2\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ with corresponding probabilities $p_1 = p_2 = 1/2$.
Show that the density matrix is in a mixed state.

3- (a) A single spin-qubit is subjected to an Hamiltonian $H = \omega\sigma_x$ with frequency $\omega = \frac{5\pi}{2} \times 10^8$ Hz.
Find the time duration τ to obtain a quantum NOT gate . (10p)



(b) Write the Hamiltonian of the three interacting qubits (with the correct order in tensor product forms). Choose flip-flop type interaction for the interaction Hamiltonian. (15p)
(J_{kl} : interaction strengths)

The questions will be graded as **:**
1 (a) – 25, 1 (b) – 25 , 2 (a) – 15, 2 (b) – 15, 3 (a) –10, 3 (b) –10

GOOD LUCK - D.T.